

full answer is $0011110100100110111011100011$

outputs match!

HW 4

1. Show that the val of x is the same, & rate as parity for all data bits, row bits, or col bits.

$$\begin{array}{c|ccc} \text{row} & b_{1,1} & \dots & b_{1,j} & r_1 \\ y=1 & b_{2,1} & \dots & b_{2,j} & r_2 \\ & \vdots & & \vdots & \\ & b_{i,1} & \dots & b_{i,j} & r_i \\ \hline & c_1 & \dots & c_j & p \end{array}$$

$$p \text{ by def} = \sum_{x=1}^i \sum_{y=1}^j b_{x,y}$$

$$p = \bigoplus_{x=1}^i \bigoplus_{y=1}^j b_{x,y}$$

$$f_y = \frac{\partial}{\partial y} \ln(x, y)$$

$$P = \frac{j}{\theta} \gamma$$

$$r_x = \bigoplus_{y=1}^j b_{x,y}$$

$$p = \frac{1}{x} \sqrt{x}$$

2. If v_1 and v_2 are codewords, then $d(v_1, v_2) = w(v_1 \oplus v_2)$, where $d(v_1, v_2)$ is the dist between them and $w(v)$ is the hamming weight of v .

v_1 and v_2 are the

a. $v_1 \oplus v_2$ returns 0 if same, 1 if diff.

$w()$ will sum these up into some number, which is exactly how the $d()$ function works

b. $d_{min} = \min d(v_1, v_2) = \min w(v_1 \oplus v_2)$.

Since $v_1 \oplus v_2 = v$, $d_{min} = \min w(v)$ so long as $v \neq 0$ which is the zero codeword

3. CRC error detection scheme, $P(x) = x^4 + x + 1$, Encode $10010011011 \rightarrow x^{10} + x^7 + x^4 + x^3 + x^1 + x^0$

a. poly of deg 4

$10010011011 + 0000$

10011

000010110

10011

0010111

100110

0010000

10011

0001100

Parading = 10010011011100

b. New string received: 00011011100

104. 000110111100

[illegible]

||| $\leftarrow$ remainder $\neq 0$, Error detected

1. new string: 00001000111100

000010101100

No remainder. Errors not detected.

4. 5 times, decides based on majority (3+), P of error = 10^{-3}

probability of error occurring 3 times in 5

$$= \sum_{k=3}^5 \binom{5}{k} \cdot (10^{-3})^k \cdot (1 - 10^{-3})^{5-k}$$

$$\binom{n}{m} = \frac{n!}{m!(n-m)!}$$

$$\begin{aligned} &< 9.985006 \times 10^{-9} \approx P \\ &\text{or } 9.985 \times 10^{-9} \times 10^9 = P \\ &\text{chance of error } 9.985 \times 10^{-9} \end{aligned}$$

5. 1-Mbps satellite link w/ prop delay 270ms. no switch delay
HDLC frames 1024 bits w/ 3-bit seq num, max possible
throughput? That is, what is throughput of data bits
carried in HDLC frames

$$\frac{1024}{1000000} = 0.001024 \text{ s} = 1.024 \text{ ms}$$

3 bit seq num means
max window of 7 frames
($2^3 - 1$) w/out ACK

$$t = 270 \times 2 + 1.024 \text{ s} = 541.024 \text{ ms}$$

Problem exists for data bps

HPLC frame:

flag	addr	ctrl	data	FCS	flag
8	8	8/16	variable	16/32	8

↑

assuming mins for ctrl, data is only

$$1024 - (8 + 8 + 8 + 16 + 8) = 976 \text{ bps} \text{ (could be less if ctrl is 16, or FCS is 32)}$$

$$\frac{976 * 7}{511,024 * 10^3} = 12,627,9086, \text{ or } \approx 12,628 \text{ bps}$$